Manan Chahal

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Summary

Fourth-year Computer Engineering student with industry experience building AI-powered applications and scalable backends at the National Stock Exchange of India. Skilled in applied ML, computer vision, multilingual AI, LLM-powered tools, and full-stack development.

Internships

AI/ML Intern – National Stock Exchange of India (NSE) May 2026 – July 2026

- Worked on an automated multilingual translation system using Sarvam AI with server-side rendering, DOM traversal, and translation batching.
- Explored Redis and Elasticsearch caching with sentence hashing and cache invalidation for translated content retrieval.
- Contributed to an internal ChatGPT-style assistant using Next.js, Redis, and RAG deployed on Linux.
- Implemented speaker diarization using PyAnnote and explored ONNX/SafeTensors conversion for model deployment.

Projects

AI-Powered Civic Issue Reporting Platform *Python, PyTorch, OpenCV, Node.js, Express.js, Leaflet.js*

- Fine-tuned a ResNet-34 model to verify complaint images before submission, reducing false reports.
- Built an interactive geospatial interface using Leaflet.js with distance-based dynamic loading of complaints.
- Developed backend APIs with Node.js and Express.js, implementing rate limiting to mitigate spam.

Cloud-Based E-Learning Platform *React, Express.js, PostgreSQL, Azure, Linux, Nginx*

- Built a full-stack web application using React and Express.js with REST APIs and auth middleware.
- Deployed on Microsoft Azure Linux VMs with SSL termination via Nginx reverse proxy.

MW Satellite Data Processing CSRE, IIT Bombay *Python, Random Forest, QGIS, PolSARpro*

- Built a Random Forest classifier to identify glacier flow lines from microwave satellite imagery.
- Processed SAR datasets using QGIS and PolSARpro to generate interpretable geospatial outputs.

Education

2023 – 2027	B.Tech in Computer Engineering at **Vidyalankar Institute of Technology**, University of Mumbai	CGPI: **9.87**
2023	Class XII (CBSE) – Green Fingers Global School	85.8%
2021	Class X (ICSE) – Ryan International School	85.4%

Achievements

- **AI/ML Intern** at NSE India
- **1st Place** CodeSprint (CSI-VIT)
- **2nd Place** Bug Auction | **3rd Place** Jeopardy (ITSA-VIT).
- Solved **350+** problems on LeetCode & competitive programming platforms.

Skills

Languages	Python, Java, JavaScript, C
AI / ML	PyTorch, Scikit-learn, OpenCV, LLMs, RAG, Speaker Diarization, ONNX
Web / DB	React, Next.js, Node.js, Express.js, Redis, PostgreSQL, MongoDB, Elasticsearch
Cloud / DevOps	Azure, Linux, Git, GitHub, Nginx, CI/CD